

# WEEK 4 – NETWORK SECURITY

Network Setup, Apache Web Server Configuration, Nmap Scanning and Wireshark Traffic Analysis

DG Interns Hub Cybersecurity Internship

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*Authorized Virtual Network Security Lab*

Tools: Kali Linux | Ubuntu | Apache | Nmap | Wireshark | VirtualBox

# 1. Introduction to Network Security

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Network security refers to the practice of protecting computer networks, systems, and data from unauthorized access, misuse, or disruption. It combines technology, processes, and policy to safeguard the confidentiality, integrity, and availability of information as it moves across a network.

Network security matters because nearly every modern service — from web applications to internal business systems — is reachable over a network, and any exposed weakness can be a path to data loss, service disruption, or unauthorized access. Two foundational practices underpin most security assessments:

- Network reconnaissance — systematically discovering which hosts are alive and reachable on a network before anything else can be assessed.
- Service enumeration — identifying exactly which services are listening on a host and on which ports, since an assessment cannot secure what it has not first identified.

Alongside these, network traffic analysis — observing the actual packets exchanged between systems — provides ground-truth visibility that scan results alone cannot: it confirms how a service really behaves when used, rather than only whether a port responds. Identifying exposed services and monitoring network traffic together form the basis of any responsible security assessment, since a service that is exposed but unmonitored is effectively invisible to the defender until something goes wrong.

## 2. Objective of the Week 4 Assignment

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The objective of this Week 4 assignment was to build a small, authorized virtual lab and use it to practise the full arc of a basic network security assessment, end to end:

- Create a small virtual network using VirtualBox.
- Configure Kali Linux as the attacker machine and Ubuntu Linux as the target machine.
- Connect both systems to the same Host-Only network and verify connectivity.
- Install and configure an Apache web server on Ubuntu, and verify HTTP access from Kali.
- Perform network scanning using Nmap to identify open ports and services, and save the scan output.
- Capture network traffic using Wireshark and analyse HTTP, ICMP, and DNS traffic.
- Document the complete network security assessment workflow.

### 3. Lab Environment

The lab was built entirely inside an isolated, authorized virtual environment for the purpose of this assignment. No public or third-party systems were involved at any stage.

| Component      | Configuration                          |
|----------------|----------------------------------------|
| Host OS        | Windows 11                             |
| Hypervisor     | Oracle VirtualBox                      |
| Attacker       | Kali Linux                             |
| Attacker IP    | 192.168.56.102                         |
| Target         | Ubuntu Linux                           |
| Target IP      | 192.168.56.101                         |
| Network        | 192.168.56.0/24 (VirtualBox Host-Only) |
| Target Service | Apache HTTP                            |
| HTTP Port      | 80                                     |
| Security Tools | Nmap, Wireshark                        |

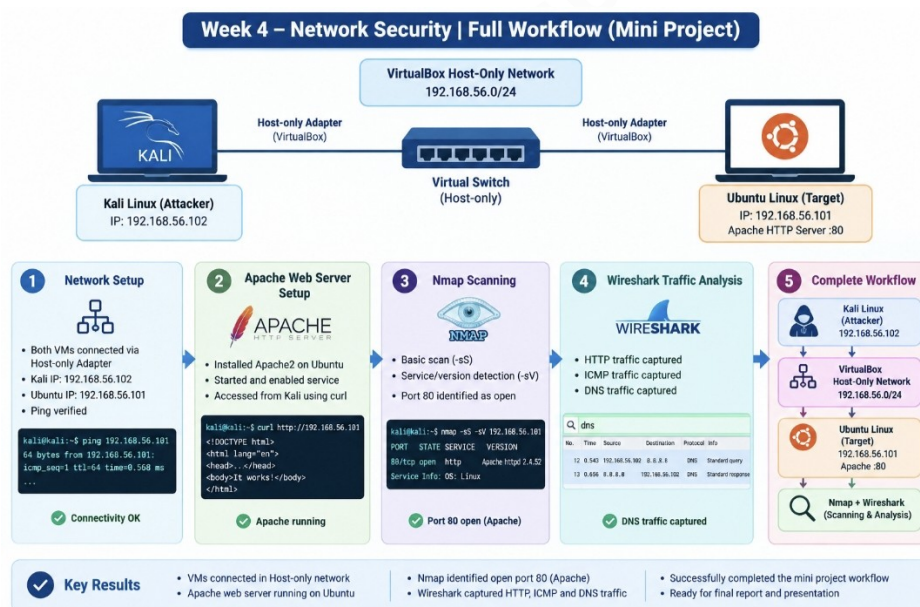


Figure 1: VirtualBox Host-Only Network and Week 4 Security Assessment Workflow

Kali (attacker) and Ubuntu (target) were connected through the same VirtualBox Host-Only network, 192.168.56.0/24, which isolates the lab from the internet and any external network. This environment was used strictly as an authorized lab for practising network security testing techniques.

## 4. Network Configuration

Kali Linux was configured as the attacker machine and Ubuntu Linux as the target machine, with both connected through the same VirtualBox Host-Only network so they could communicate directly and in isolation from any external network.

- Kali IP: 192.168.56.102
- Ubuntu IP: 192.168.56.101
- Network: 192.168.56.0/24
- Connectivity was verified using ICMP ping from Kali to Ubuntu.
- The environment was isolated and authorized for testing.

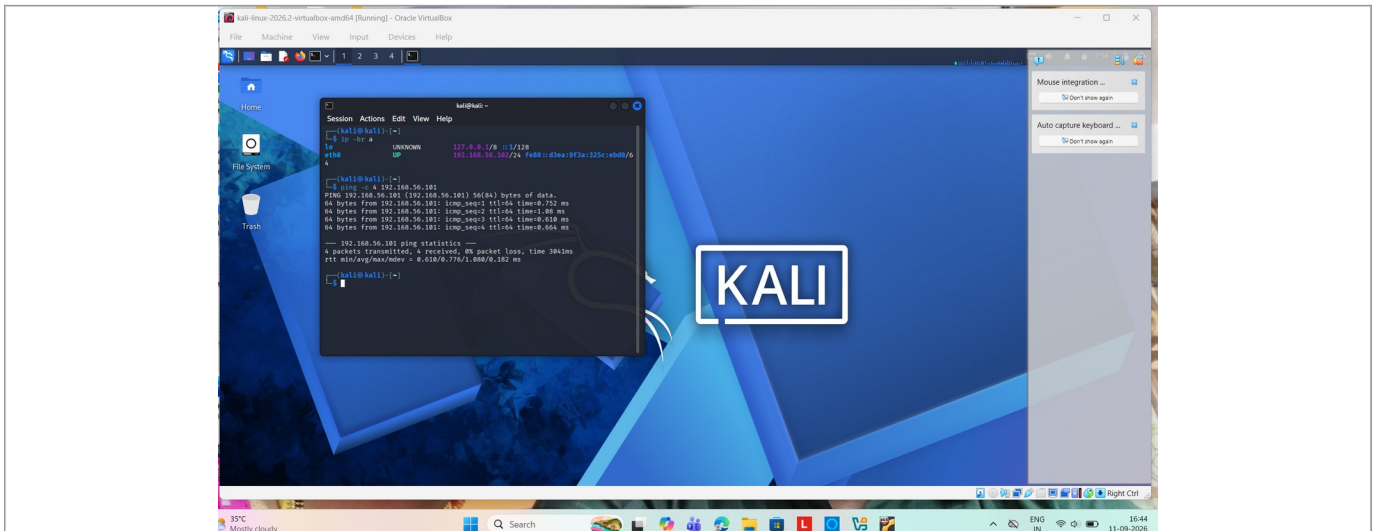


Figure 2: Kali and Ubuntu Network Configuration with Successful Connectivity Verification

The Kali terminal output confirms the eth0 interface address of 192.168.56.102/24, followed by a successful ping test to 192.168.56.101 with 0% packet loss — confirming that both machines could reach each other over the Host-Only network before any further testing began.

## 5. Apache Web Server Configuration

Apache2 was installed on the Ubuntu target machine, enabled, and started as a system service. Its status was verified directly on Ubuntu, and HTTP access was then confirmed from Kali.

**Target:** 192.168.56.101

**HTTP:** TCP port 80

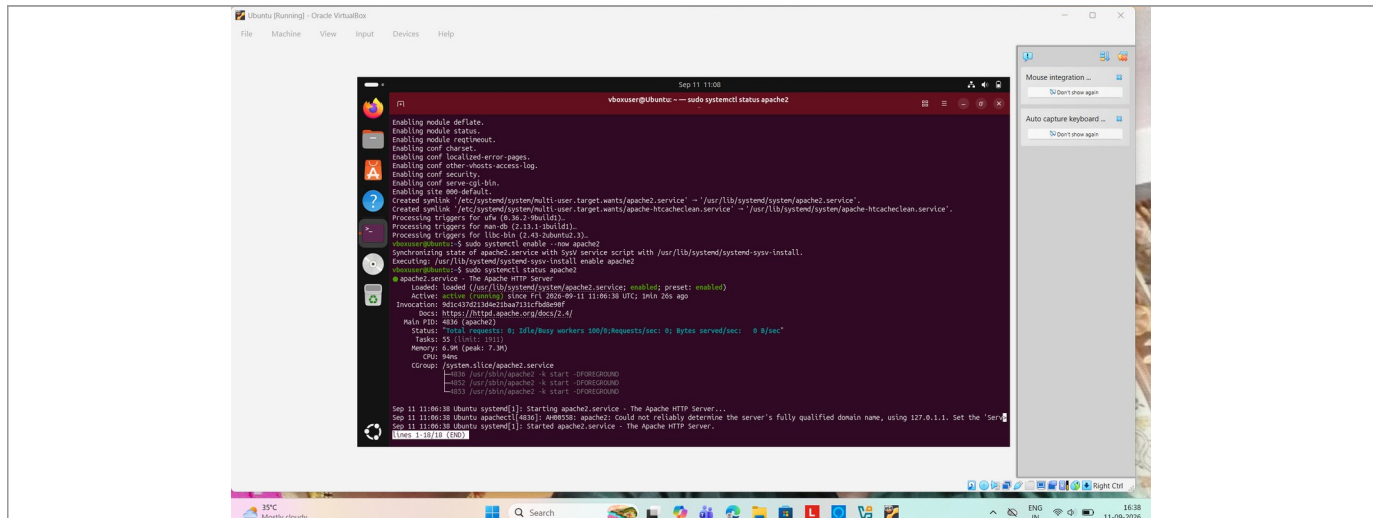


Figure 3: Apache2 Web Server Running on Ubuntu

The systemctl status apache2 output confirms the service was loaded, enabled, and active (running) on Ubuntu.

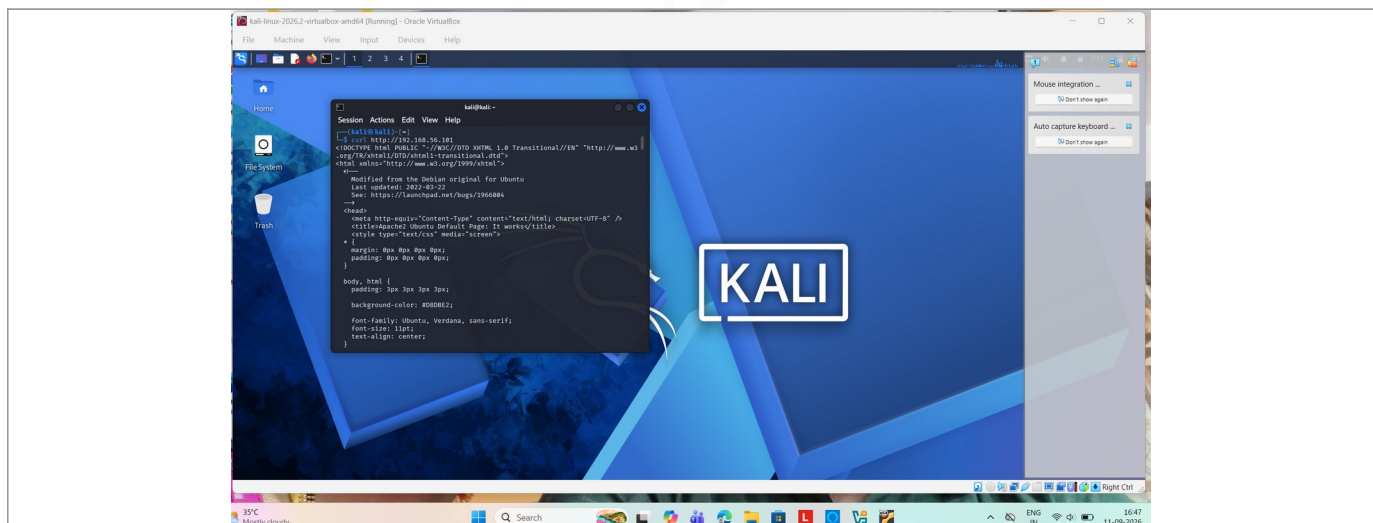


Figure 4: Apache Web Server Accessed from Kali Using curl

From Kali, curl http://192.168.56.101 was used to request the default Apache page directly over the Host-Only network. The successful HTML output returned by curl (the standard Apache2 Ubuntu default page) confirmed application-layer connectivity between Kali and the Ubuntu Apache web server, not just basic network reachability.

## 6. Nmap Network Scanning

Nmap was used from Kali to assess the Ubuntu target and identify which network services were reachable.

**Target scanned:** 192.168.56.101

**nmap** 192.168.56.101

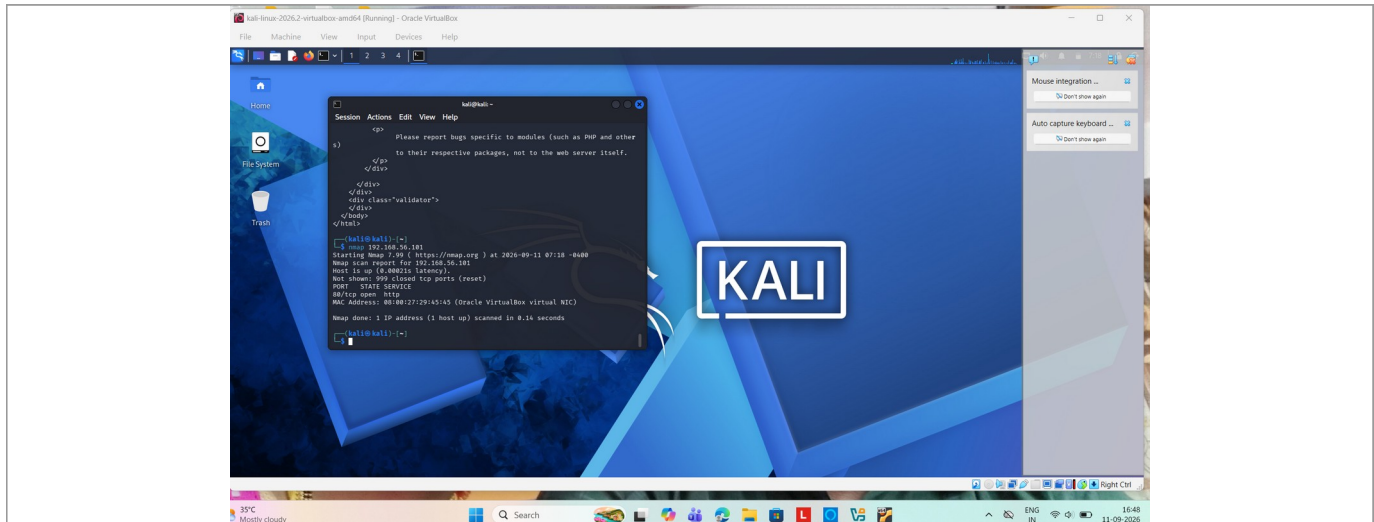


Figure 5: Basic Nmap Scan of the Ubuntu Target

The scan confirmed the target was up and reachable, with the following result:

**80/tcp open http**

This indicates that the HTTP service was reachable on TCP port 80, consistent with the Apache server configured in Task 2.

*An open port is not, by itself, a confirmed vulnerability. TCP port 80 being open simply means the HTTP service is reachable as intended; exposed services like this should be securely configured, monitored, and limited to required functionality rather than treated as a finding on their own.*

## 7. Nmap Service and Version Detection

```
nmap -sV 192.168.56.101
```

Nmap's service/version detection mode was used to gather additional information about the service running on the open port, beyond simply confirming that it was open.

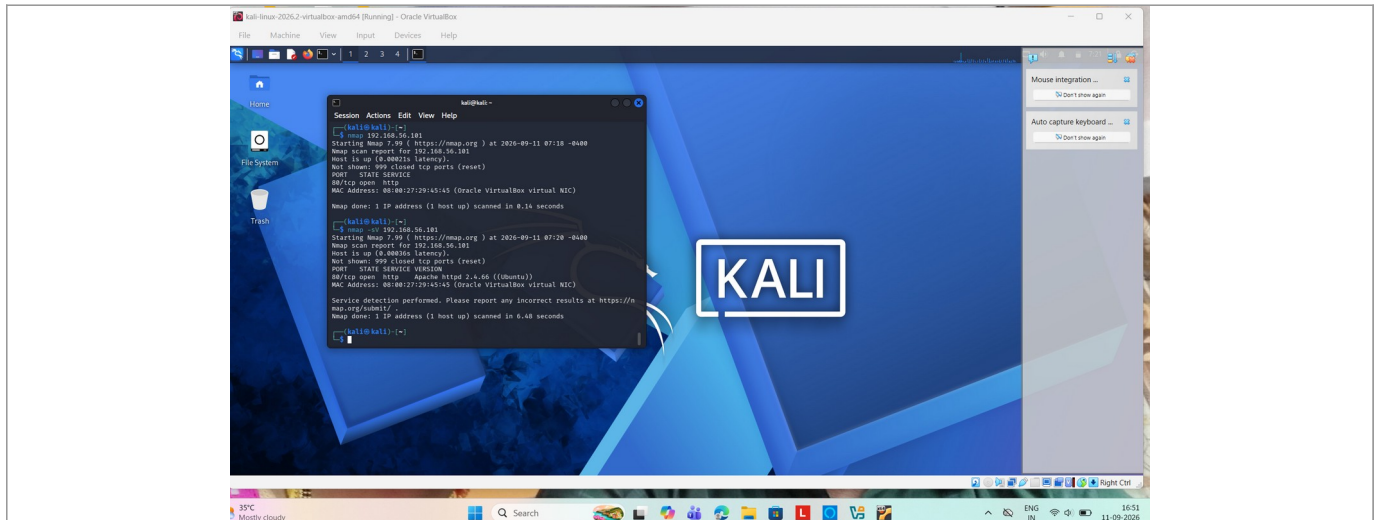


Figure 6: Nmap Service and Version Detection

The scan reported port 80 as open, running the http service, identified as Apache httpd 2.4.66 ((Ubuntu)) — this version string was clearly visible in the scan output shown above.

**Saved scan output:** week4\_nmap\_scan.txt

The scan output was saved to a text file to maintain a reproducible record of the assessment for documentation and later analysis, alongside the screenshots captured for this report.

## 8. HTTP Traffic Analysis

Wireshark was used on Kali for packet-level network traffic analysis. HTTP communication was generated by accessing the Ubuntu Apache server from Kali:

```
curl http://192.168.56.101
```

Traffic on TCP port 80 was captured using the following Wireshark display filter:

```
tcp.port == 80
```

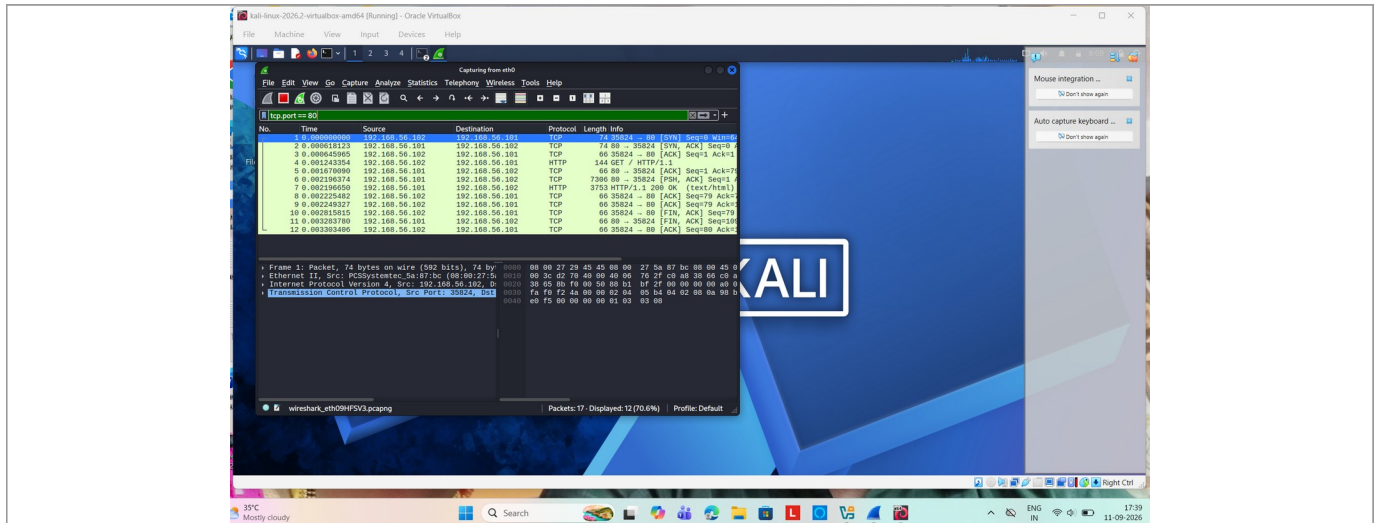


Figure 7: HTTP Traffic Captured Between Kali and Ubuntu

The capture shows the full TCP/HTTP exchange between Kali and Ubuntu: a three-way handshake (SYN, SYN/ACK, ACK), the HTTP GET request, the HTTP/1.1 200 OK response carrying the page content, and the connection teardown (FIN/ACK exchanges).

### Observations

- Kali acted as the HTTP client (source port 35824 in the capture).
- Ubuntu acted as the HTTP server, listening on TCP port 80.
- The packet capture provided full visibility into the request/response exchange at the application layer, not just the fact that a connection succeeded.

## 9. ICMP Traffic Analysis

ICMP traffic was generated using ping from Kali toward Ubuntu:

```
ping -c 4 192.168.56.101
```

Kali sent ICMP Echo Request packets to Ubuntu, and Ubuntu responded with ICMP Echo Reply packets for each one — verifying network-level reachability and demonstrating how ICMP traffic appears in a packet capture.

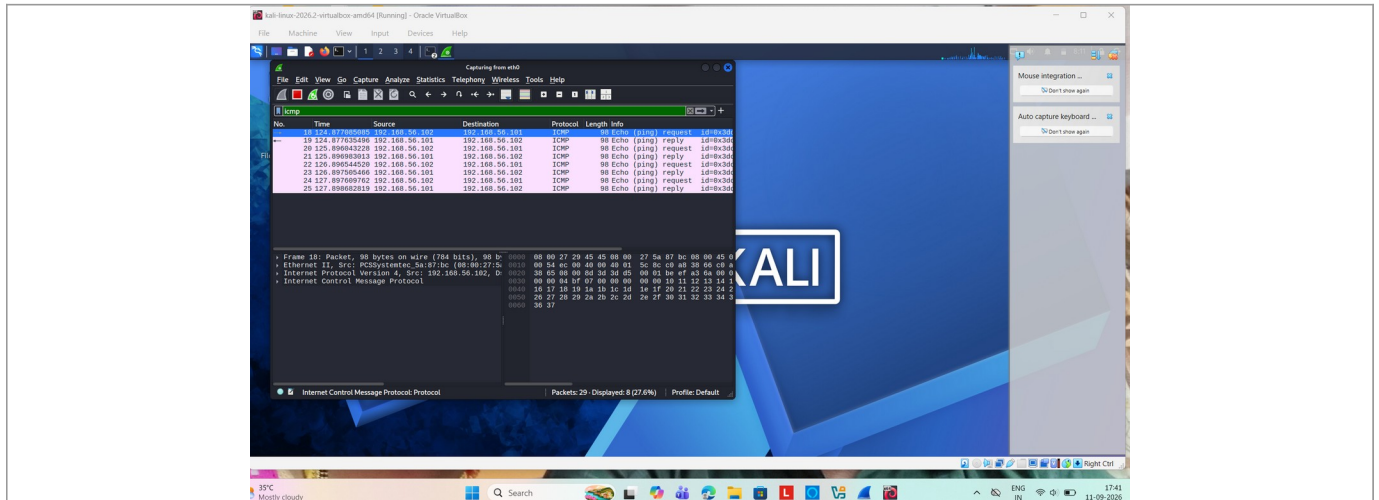


Figure 8: ICMP Echo Request and Echo Reply Traffic

## 10. DNS Traffic Analysis

DNS traffic was captured separately using a temporary NAT connection on Kali, since the Host-Only network itself does not provide internet or DNS access. DNS resolution was generated using:

```
nslookup google.com
```

Wireshark was used to observe the DNS query and response traffic, using the display filter:

dns

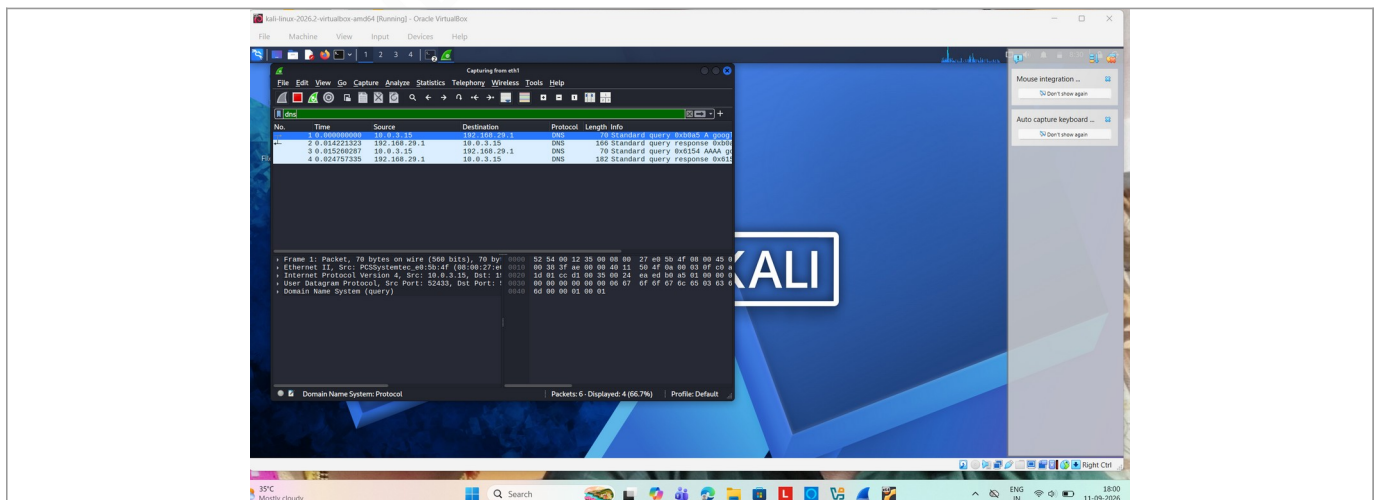


Figure 9: DNS Query and Response Traffic Captured in Wireshark

This DNS capture used Kali's temporary NAT interface (eth1, IP 10.0.3.15) rather than the Host-Only lab network, because

*the Host-Only segment has no route to the internet and therefore no DNS resolution path. This traffic is not part of the Kali-to-Ubuntu Host-Only communication. After the DNS capture, Kali was restored to its Host-Only lab configuration for the remainder of the assessment.*

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## 11. Complete Mini-Project Workflow

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The assignment followed this actual sequence of steps:

- 1. Created the virtual lab.
- 2. Configured Kali as attacker.
- 3. Configured Ubuntu as target.
- 4. Verified connectivity.
- 5. Installed and configured Apache.
- 6. Tested HTTP access from Kali.
- 7. Performed Nmap scanning.
- 8. Identified TCP port 80 / HTTP.
- 9. Captured HTTP traffic.
- 10. Captured ICMP traffic.
- 11. Captured DNS traffic.
- 12. Documented the findings.

## 12. Security Observations

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| Observation                    | Security Significance                                                                |
|--------------------------------|--------------------------------------------------------------------------------------|
| Apache exposed on TCP/80       | Web services should be securely configured, monitored, and appropriately restricted. |
| Network connectivity confirmed | Demonstrated successful communication between the authorized lab machines.           |
| Nmap identified HTTP service   | Demonstrates the importance of service enumeration during network assessment.        |
| HTTP traffic captured          | Demonstrates visibility into application-layer network traffic.                      |
| ICMP traffic captured          | Demonstrates network-level connectivity and packet analysis.                         |
| DNS traffic captured           | Demonstrates DNS request/response analysis.                                          |

*These are observations and security considerations from the lab, not confirmed vulnerabilities.*

## 13. Security Best Practices

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- Expose only required services.
- Keep web servers updated.
- Use firewall rules to restrict unnecessary access.
- Monitor network traffic.
- Perform regular service enumeration.
- Prefer HTTPS over unencrypted HTTP in real-world environments.
- Restrict administrative services.
- Maintain logs and monitor suspicious activity.
- Use isolated environments for authorized security testing.

## 14. Learning Outcomes

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This assignment provided hands-on practice with the following skills:

- Virtual network configuration
- Kali Linux
- Ubuntu Linux
- Apache configuration
- Nmap reconnaissance
- Service enumeration
- Wireshark packet capture
- HTTP analysis
- ICMP analysis
- DNS analysis
- Basic network security assessment
- Security documentation

## 15. Conclusion

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The Week 4 assignment provided hands-on experience with network setup, web service deployment, network reconnaissance, service discovery, packet capture, protocol analysis, and basic security assessment — moving through each stage using a real, authorized virtual lab rather than a purely theoretical exercise.

The lab demonstrated the complete workflow:

**Network Setup → Service Deployment → Scanning → Packet Capture → Traffic Analysis → Security Observations → Documentation**

## 16. Tools Used

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- Oracle VirtualBox
- Kali Linux
- Ubuntu Linux
- Apache2
- Nmap
- Wireshark
- curl
- ping
- nslookup

## 17. References

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- Nmap Documentation — <https://nmap.org/docs.html>
- Wireshark Documentation — <https://www.wireshark.org/docs/>
- Apache HTTP Server Documentation — <https://httpd.apache.org/docs/>
- Kali Linux Documentation — <https://www.kali.org/docs/>
- Ubuntu Documentation — <https://help.ubuntu.com/>
- Oracle VirtualBox Documentation — <https://www.virtualbox.org/wiki/Documentation>